Coloured Filters Investigation

Aim: to observe how the colours we percieve are changed by filters.

Equipment

- ray boxes
- coloured filters
- lab pack (if needed)
- coloured modelling clay (see results table)

Safety Information

Filament bulb – risk of burns.

Handle with extra care. Rinse any burns under cold running water for 10 minutes and inform the teacher immediately.

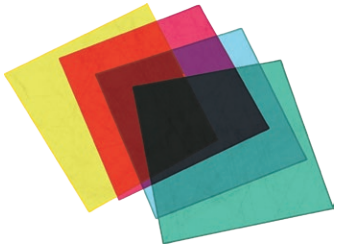
Method

Step 1: Set up the ray box and place the modelling clay into the light beam.

Step 2: place a coloured filter into the slot.

Step 3: Observe the colour of the clay under the light and record the results.

Step 4: Repeat steps 1 to 3 for each combination of filter(s) and coloured clay.

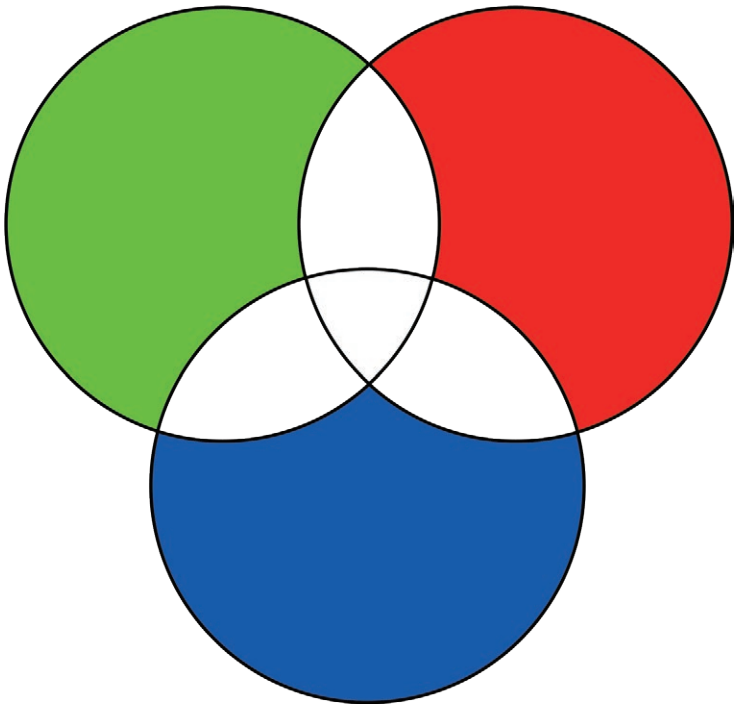


Results

Colour of Filter(s)	Colour of Modelling Clay			
	white	red	blue	green
none				
red				
blue				
green				
yellow				
magenta				
cyan				

Questions

1. Draw the spectrum of white light as seen through a prism.
2. Name the three primary colours of light.
3. Name the three secondary colours of light.
4. Explain how colour is perceived in terms of absorption and reflection.
5. Explain how a filter works and how it makes an object appear a different colour.
6. Copy and complete the diagram to show the colours which make up white light.

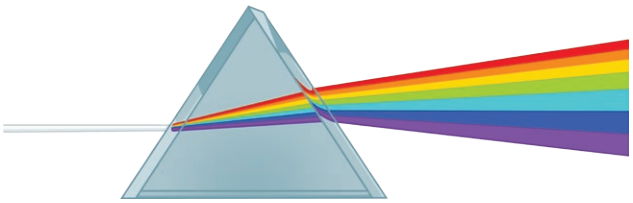


Answers

Results

Colour of Filter(s)	Colour of Modelling Clay			
	white	red	blue	green
none	**white**	**red**	**blue**	**green**
red	**red**	**red**	**black**	**black**
blue	**blue**	**black**	**blue**	**black**
green	**green**	**black**	**black**	**green**
yellow	**yellow**	**red**	**black**	**green**
magenta	**magenta**	**red**	**blue**	**black**
cyan	**cyan**	**black**	**blue**	**green**

Questions

1. 

2. Red, blue and green.

3. Yellow, magenta and cyan.

4. The colour of the object is the one which is reflected into our eye. Any other colours are absorbed by the object.

5. A filter only allows that colour to pass through it. If the object is a different colour to the light being filtered through, it will appear black.

6. 